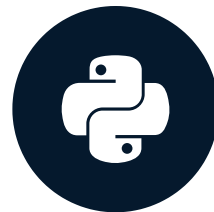


Grouping and summing: the beginner's pivot table

PYTHON FOR SPREADSHEET USERS



Chris Cardillo
Data Scientist

Fruit stores

	A	B	C	D
1	store	product_name	quantity_purchased	revenue
2	Pete's Discount Fruit	Banana	1	0.23
3	Derek's Fruit Stand	Banana	3	0.69
4	Pete's Discount Fruit	Orange	1	0.68
5	Derek's Fruit Stand	Orange	2	1.36
6	Pete's Discount Fruit	Apple	1	0.88
7	Derek's Fruit Stand	Apple	1	0.88
8	Derek's Fruit Stand	Plum	1	0.96
9	Derek's Fruit Stand	Plum	2	1.92
10	Derek's Fruit Stand	Kiwi	2	2.24
11	Derek's Fruit Stand	Watermelon	1	3.98

Fruit stores

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1	store	product_name	quantity_purchased	revenue
2	Pete's Discount Fruit	Banana	1	0.23
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7	Derek's Fruit Stand	Apple	1	0.88
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9	Derek's Fruit Stand	Plum	2	1.92
10	Derek's Fruit Stand	Kiwi	2	2.24
11	Derek's Fruit Stand	Watermelon	1	3.98

Fruit stores

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Fruit stores

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9	Derek's Fruit Stand	Plum	2	1.92
10	Derek's Fruit Stand	Kiwi	2	2.24
11	Derek's Fruit Stand	Watermelon	1	3.98

Pivoting in spreadsheets

RAW DATA

	A	B	C	D
1	store	product_name	quantity_purchased	revenue
2	Pete's Discount Fruit	Banana	1	0.23
3	Derek's Fruit Stand	Banana	3	0.69
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9	Derek's Fruit Stand	Plum	2	1.92
10	Derek's Fruit Stand	Kiwi	2	2.24
11	Derek's Fruit Stand	Watermelon	1	3.98

PIVOT TABLE

	A	B
1	SUM of quantity_purchased	SUM of revenue
2	57	133.12

PIVOT TABLE EDITOR

Pivot table editor

Sheet1!A1:D33

Suggested

RowsADD

ColumnsADD

Values as: ColumnsADD

quantity_purchased

Summarize bySUMShow asDefault

revenue

Summarize bySUMShow asDefault

FiltersADD

Pivoting in spreadsheets

RAW DATA

	A	B	C	D
1	store	product_name	quantity_purchased	revenue
2	Pete's Discount Fruit	Banana	1	0.23
3	Derek's Fruit Stand	Banana	3	0.69
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PIVOT TABLE

	A	B
1	SUM of quantity_purchased	SUM of revenue
2	57	133.12

PIVOT TABLE EDITOR

Pivot table editor

Sheet1!A1:D33

Suggested

RowsADD

ColumnsADD

Values as: ColumnsADD

quantity_purchased

Summarize bySUMShow asDefault

revenue

Summarize bySUMShow asDefault

FiltersADD

.sum()

```
fruit_sales.head()
```

		store	product_name	quantity_purchased	revenue
0	Pete's Discount	Fruit	Banana	1	0.23
1	Derek's Fruit	Stand	Banana	3	0.69
2	Pete's Discount	Fruit	Orange	1	0.68
3	Derek's Fruit	Stand	Orange	2	1.36
4	Pete's Discount	Fruit	Apple	1	0.88

```
fruit_sales.sum(numeric_only=True)
```

quantity_purchased	57.00
revenue	133.12
dtype:	float64

Pure summary with .sum()

IN SPREADSHEETS

	A	B
1	SUM of quantity_purchased	SUM of revenue
2	57	133.12

IN PYTHON

```
fruit_sales.sum(numeric_only=True)
```

```
quantity_purchased    57.00  
revenue                133.12  
dtype: float64
```

Using pivot table rows in spreadsheets

PIVOT TABLE

	A	B	C
1	store	SUM of quantity_purchased	SUM of revenue
2	Derek's Fruit Stand	27	57.22
3	Pete's Discount Fruit	30	75.9

PIVOT TABLE EDITOR

Pivot table editor

Sheet1!A1:D33

Suggested

Rows

store

Order
Ascending

Sort by
store

☐ Show totals

Columns

Values as: Columns

quantity_purchased

Summarize by
SUM

Show as
Default

revenue

Summarize by
SUM

Show as
Default

Filters

A simple pivot table in Python - .groupby().sum()

In Python

```
fruit_sales.groupby('store', as_index=False).sum()
```

	store	quantity_purchased	revenue
0	Derek's Fruit Stand	27	57.22
1	Pete's Discount Fruit	30	75.90

In spreadsheets

	A	B	C
1	store	SUM of quantity_purchased	SUM of revenue
2	Derek's Fruit Stand	27	57.22
3	Pete's Discount Fruit	30	75.9

A simple pivot table in Python - .groupby().sum()

In Python

```
fruit_sales.groupby('store', as_index=False).sum()
```

IN SPREADSHEETS PIVOT TABLE EDITOR

Rows

ADD

store

X

Order

Ascending

▼

Sort by

store

▼

☐ Show totals

Columns

ADD

Values as: Columns

▼

ADD

quantity_purchased

X

Summarize by

SUM

▼

Show as

Default

▼

revenue

X

Summarize by

SUM

▼

Show as

Default

▼

A simple pivot table in Python - .groupby().sum()

In Python

```
fruit_sales.groupby('store', as_index=False).sum()
```

	store	quantity_purchased	revenue
0	Derek's Fruit Stand	27	57.22
1	Pete's Discount Fruit	30	75.90

In spreadsheets

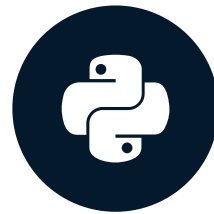
	A	B	C
1	store	SUM of quantity_purchased	SUM of revenue
2	Derek's Fruit Stand	27	57.22
3	Pete's Discount Fruit	30	75.9

Your turn!

PYTHON FOR SPREADSHEET USERS

Grouping by multiple columns

PYTHON FOR SPREADSHEET USERS



Chris Cardillo
Data Scientist

Fruit sales

```
fruit_sales
```

		store	product_name	quantity_purchased	revenue
0	Pete's Discount	Fruit	Banana	1	0.23
1	Derek's Fruit	Stand	Banana	3	0.69
2	Pete's Discount	Fruit	Orange	1	0.68
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9	Derek's Fruit	Stand	Watermelon	1	3.98
10	Pete's Discount	Fruit	Blueberries	3	15.48
11	Derek's Fruit	Stand	Dragonfruit	3	15.81
12	Pete's Discount	Fruit	Banana	1	0.23
13	Derek's Fruit	Stand	Banana	1	0.23
14	Pete's Discount	Fruit	Apple	3	2.64
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16	Pete's Discount	Fruit	Kiwi	3	3.36
17	Derek's Fruit	Stand	Kiwi	3	3.36
18	Derek's Fruit	Stand	Watermelon	2	7.96
19	Pete's Discount	Fruit	Blueberries	3	15.48
20	Derek's Fruit	Stand	Blueberries	1	5.16
21	Derek's Fruit	Stand	Blueberries	1	5.16
22	Pete's Discount	Fruit	Dragonfruit	3	15.81
23	Pete's Discount	Fruit	Dragonfruit	1	5.27

```
fruit_sales.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 32 entries, 0 to 31  
Data columns (total 4 columns):  
store                32 non-null object  
product_name         32 non-null object  
quantity_purchased   32 non-null int64  
revenue              32 non-null float64  
dtypes: float64(1), int64(1), object(2)  
memory usage: 1.1+ KB
```

Fruit sales

```
fruit_sales
```

		store	product_name	quantity_purchased	revenue
0	Pete's Discount	Fruit	Banana	1	0.23
1	Derek's Fruit	Stand	Banana	3	0.69
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dtypes: float64(1), int64(1), object(2)  
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```

Fruit sales

		store	product_name	quantity_purchased	revenue
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6	Derek's Fruit	Stand	Plum	1	0.96
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10	Pete's Discount	Fruit	Blueberries	3	15.48
11	Derek's Fruit	Stand	Dragonfruit	3	15.81
12	Pete's Discount	Fruit	Banana	1	0.23
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20	Derek's Fruit	Stand	Blueberries	1	5.16
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22	Pete's Discount	Fruit	Dragonfruit	3	15.81
23	Pete's Discount	Fruit	Dragonfruit	1	5.27

Adding a list of column names

BEFORE

```
fruit_sales.groupby('store', as_index=False).sum()
```

AFTER

```
fruit_sales.groupby(['store', 'product_name'], as_index=False).sum()
```

What is a list?

```
shopping_list = ['milk', 'eggs', 'cheese']
```

By store, by fruit

```
groups = ['store', 'product_name']
```

```
fruit_sales_less = fruit_sales.groupby(groups, as_index=False).sum()
```

	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
1	Derek's Fruit Stand	Banana	4	0.92
2	Derek's Fruit Stand	Blueberries	2	10.32
3	Derek's Fruit Stand	Dragonfruit	4	21.08
4	Derek's Fruit Stand	Kiwi	5	5.60
5	Derek's Fruit Stand	Orange	4	2.72
6	Derek's Fruit Stand	Plum	3	2.88
7	Derek's Fruit Stand	Watermelon	3	11.94
8	Pete's Discount Fruit	Apple	4	3.52
9	Pete's Discount Fruit	Banana	2	0.46
10	Pete's Discount Fruit	Blueberries	6	30.96
11	Pete's Discount Fruit	Dragonfruit	4	21.08
12	Pete's Discount Fruit	Kiwi	6	6.72
13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

By store, by fruit

```
groups = ['store', 'product_name']
```

```
fruit_sales_less = fruit_sales.groupby(groups, as_index=False).sum()
```

	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
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13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

The benefits of grouping by more columns before `.sum()`

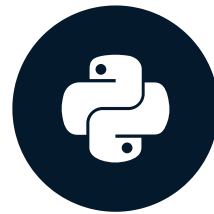
- It's not "one or none"
- Reduce data down to what matters
- Help make spreadsheet data more manageable

Your turn!

PYTHON FOR SPREADSHEET USERS

More useful summary tools

PYTHON FOR SPREADSHEET USERS



Chris Cardillo
Data Scientist

.mean()

```
fruit_sales.groupby('store', as_index=False).mean()
```

	store	quantity_purchased	revenue
0	Derek's Fruit Stand	1.6875	3.57625
1	Pete's Discount Fruit	1.8750	4.74375

Steps to an answer

1. Fruit store transactions
2. Total sales by fruit for each store
3. Sorted sales by fruit for each store
4. Top row for each store

1: Fruit store transactions

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Banana	1	0.23
1	Derek's Fruit Stand	Banana	3	0.69
2	Pete's Discount Fruit	Orange	1	0.68
3	Derek's Fruit Stand	Orange	2	1.36
4	Pete's Discount Fruit	Apple	1	0.88
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13	Derek's Fruit Stand	Banana	1	0.23
14	Pete's Discount Fruit	Apple	3	2.64
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19	Pete's Discount Fruit	Blueberries	3	15.48
20	Derek's Fruit Stand	Blueberries	1	5.16
21	Derek's Fruit Stand	Blueberries	1	5.16
22	Pete's Discount Fruit	Dragonfruit	3	15.81
23	Pete's Discount Fruit	Dragonfruit	1	5.27

2: Total sales by fruit for each store

```
totals = fruit_sales.groupby(['store', 'product_name'], as_index=False).sum()
```

	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
1	Derek's Fruit Stand	Banana	4	0.92
2	Derek's Fruit Stand	Blueberries	2	10.32
3	Derek's Fruit Stand	Dragonfruit	4	21.08
4	Derek's Fruit Stand	Kiwi	5	5.60
5	Derek's Fruit Stand	Orange	4	2.72
6	Derek's Fruit Stand	Plum	3	2.88
7	Derek's Fruit Stand	Watermelon	3	11.94
8	Pete's Discount Fruit	Apple	4	3.52
9	Pete's Discount Fruit	Banana	2	0.46
10	Pete's Discount Fruit	Blueberries	6	30.96
11	Pete's Discount Fruit	Dragonfruit	4	21.08
12	Pete's Discount Fruit	Kiwi	6	6.72
13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

3: Sorted sales by fruit for each store

```
totals = (totals.sort_values('revenue', ascending=False)
.reset_index(drop=True))
```

		store	product_name	quantity_purchased	revenue
0	Pete's Discount	Fruit	Blueberries	6	30.96
1	Derek's Fruit	Stand	Dragonfruit	4	21.08
2	Pete's Discount	Fruit	Dragonfruit	4	21.08
3	Derek's Fruit	Stand	Watermelon	3	11.94
4	Derek's Fruit	Stand	Blueberries	2	10.32
5	Pete's Discount	Fruit	Watermelon	2	7.96
6	Pete's Discount	Fruit	Kiwi	6	6.72
7	Derek's Fruit	Stand	Kiwi	5	5.60
8	Pete's Discount	Fruit	Plum	4	3.84
9	Pete's Discount	Fruit	Apple	4	3.52
10	Derek's Fruit	Stand	Plum	3	2.88
11	Derek's Fruit	Stand	Orange	4	2.72
12	Derek's Fruit	Stand	Apple	2	1.76
13	Pete's Discount	Fruit	Orange	2	1.36
14	Derek's Fruit	Stand	Banana	4	0.92
15	Pete's Discount	Fruit	Banana	2	0.46

4: Top row for each store

```
top_store_sellers = totals.groupby('store').head(1).reset_index(drop=True)
```

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Blueberries	6	30.96
1	Derek's Fruit Stand	Dragonfruit	4	21.08

Steps to an answer

```
# Raw data
fruit_sales

# Summary - by fruit by store
totals = fruit_sales.groupby(['store', 'product_name'], as_index=False).sum()

# Sort the Summary
totals = (totals.sort_values('revenue', ascending=False)
         .reset_index(drop=True))

# First row for each store
top_store_sellers = totals.groupby('store').head(1).reset_index(drop=True)
```

		store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Blueberries	6	30.96	
1	Derek's Fruit Stand	Dragonfruit	4	21.08	

Steps to an answer

1: RAW TRANSACTION DATA

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Banana	1	0.23
1	Derek's Fruit Stand	Banana	3	0.69
2	Pete's Discount Fruit	Orange	1	0.68
3	Derek's Fruit Stand	Orange	2	1.36
4	Pete's Discount Fruit	Apple	1	0.88
5	Derek's Fruit Stand	Apple	1	0.88
6	Derek's Fruit Stand	Plum	1	0.96
7	Derek's Fruit Stand	Plum	2	1.92
8	Derek's Fruit Stand	Kiwi	2	2.24
9	Derek's Fruit Stand	Watermelon	1	3.98
10	Pete's Discount Fruit	Blueberries	3	15.48
11	Derek's Fruit Stand	Dragonfruit	3	15.81
12	Pete's Discount Fruit	Banana	1	0.23
13	Derek's Fruit Stand	Banana	1	0.23
14	Pete's Discount Fruit	Apple	3	2.64
15	Pete's Discount Fruit	Plum	2	1.92
16	Pete's Discount Fruit	Kiwi	3	3.36
17	Derek's Fruit Stand	Kiwi	3	3.36
18	Derek's Fruit Stand	Watermelon	2	7.96
19	Pete's Discount Fruit	Blueberries	3	15.48
20	Derek's Fruit Stand	Blueberries	1	5.16
21	Derek's Fruit Stand	Blueberries	1	5.16
22	Pete's Discount Fruit	Dragonfruit	3	15.81
23	Pete's Discount Fruit	Dragonfruit	1	5.27

2: BY FRUIT/STORE SUMMARY

	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
1	Derek's Fruit Stand	Banana	4	0.92
2	Derek's Fruit Stand	Blueberries	2	10.32
3	Derek's Fruit Stand	Dragonfruit	4	21.08
4	Derek's Fruit Stand	Kiwi	5	5.60
5	Derek's Fruit Stand	Orange	4	2.72
6	Derek's Fruit Stand	Plum	3	2.88
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12	Pete's Discount Fruit	Kiwi	6	6.72
13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

Steps to an answer

2: BY FRUIT/STORE SUMMARY

	store	product_name	quantity_purchased	revenue
0	Derek's Fruit Stand	Apple	2	1.76
1	Derek's Fruit Stand	Banana	4	0.92
2	Derek's Fruit Stand	Blueberries	2	10.32
3	Derek's Fruit Stand	Dragonfruit	4	21.08
4	Derek's Fruit Stand	Kiwi	5	5.60
5	Derek's Fruit Stand	Orange	4	2.72
6	Derek's Fruit Stand	Plum	3	2.88
7	Derek's Fruit Stand	Watermelon	3	11.94
8	Pete's Discount Fruit	Apple	4	3.52
9	Pete's Discount Fruit	Banana	2	0.46
10	Pete's Discount Fruit	Blueberries	6	30.96
11	Pete's Discount Fruit	Dragonfruit	4	21.08
12	Pete's Discount Fruit	Kiwi	6	6.72
13	Pete's Discount Fruit	Orange	2	1.36
14	Pete's Discount Fruit	Plum	4	3.84
15	Pete's Discount Fruit	Watermelon	2	7.96

3: BY FRUIT/STORE SORTED

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Blueberries	6	30.96
1	Derek's Fruit Stand	Dragonfruit	4	21.08
2	Pete's Discount Fruit	Dragonfruit	4	21.08
3	Derek's Fruit Stand	Watermelon	3	11.94
4	Derek's Fruit Stand	Blueberries	2	10.32
5	Pete's Discount Fruit	Watermelon	2	7.96
6	Pete's Discount Fruit	Kiwi	6	6.72
7	Derek's Fruit Stand	Kiwi	5	5.60
8	Pete's Discount Fruit	Plum	4	3.84
9	Pete's Discount Fruit	Apple	4	3.52
10	Derek's Fruit Stand	Plum	3	2.88
11	Derek's Fruit Stand	Orange	4	2.72
12	Derek's Fruit Stand	Apple	2	1.76
13	Pete's Discount Fruit	Orange	2	1.36
14	Derek's Fruit Stand	Banana	4	0.92
15	Pete's Discount Fruit	Banana	2	0.46

Steps to an answer

3: BY FRUIT/STORE SORTED

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Blueberries	6	30.96
1	Derek's Fruit Stand	Dragonfruit	4	21.08
2	Pete's Discount Fruit	Dragonfruit	4	21.08
3	Derek's Fruit Stand	Watermelon	3	11.94
4	Derek's Fruit Stand	Blueberries	2	10.32
5	Pete's Discount Fruit	Watermelon	2	7.96
6	Pete's Discount Fruit	Kiwi	6	6.72
7	Derek's Fruit Stand	Kiwi	5	5.60
8	Pete's Discount Fruit	Plum	4	3.84
9	Pete's Discount Fruit	Apple	4	3.52
10	Derek's Fruit Stand	Plum	3	2.88
11	Derek's Fruit Stand	Orange	4	2.72
12	Derek's Fruit Stand	Apple	2	1.76
13	Pete's Discount Fruit	Orange	2	1.36
14	Derek's Fruit Stand	Banana	4	0.92
15	Pete's Discount Fruit	Banana	2	0.46

4: FIRST ROW PER STORE

	store	product_name	quantity_purchased	revenue
0	Pete's Discount Fruit	Blueberries	6	30.96
1	Derek's Fruit Stand	Dragonfruit	4	21.08

Your turn!

PYTHON FOR SPREADSHEET USERS